English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021. Name:Student no.: *Your grade is 1 + the number of points. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$. *Define* $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. *Unless otherwise stated,* $n, k, i \in \mathbb{N}$.

Problem 1. An attacker sends a phishing email to a lecturer. At a random time $X \sim U(0,1)$ during the next day, the attacker hides all traces but also loses the opportunity to benefit. Thus $f_X(x) = 1$ on $[0,1]$ and 0 elsewhere. The lecturer responds after an independent exponential time with mean two days: $Y \sim \text{Exp}(1/2)$, with $f_Y(y) = \frac{1}{2}e^{-y/2}$ for $y \geq 0$ and 0 elsewhere.

1. (0.75 points) What is the probability that the attack succeeds? Approximately what fraction of many attempts will succeed?

   Solution. The desired probability is

   $$p := \mathbb{P}(X < Y) = \int_0^1 \int_0^x \frac{1}{2} e^{-y/2}\, dy\, dx,$$

   which is easily calculated. By the law of large numbers, the successful fraction is p; specify the variables and verify the hypotheses.

2. (0.25 points) What is the probability the attacker benefits given that the lecturer responds after exactly 12 hours?

   Solution. We seek $\mathbb{P}(Y < X \mid X = 1/2) = \mathbb{P}(Y < 1/2)$. The intention was to use a conditional density, but it is actually unnecessary.

Problem 2. Two faces of a fair die are white, two green and two red. Roll it twice. Let X count white results and Y red results. Find:

1. (0.5 points) the joint distribution and correlation of X, Y. Are they independent?
2. (0.25 points) $\mathbb{P}(X = 1 \mid Y = 1)$;
3. (0.25 points) $\mathbb{P}(X > Y)$.

Solution in the source: —

Problem 3. Daily accident counts in Bulgaria are modelled by independent Poisson variables $X_1, X_2, \ldots$ with $\lambda = 10$, so $\mathbb{P}(X_1 = k) = e^{-\lambda}\lambda^k/k!$ for $k \in \mathbb{N}_0$. Set $S_n = X_1 + \cdots + X_n$.

1. (0.25 points) Find $\mathbb{E}S_n$ and DS_n.

   Solution. $\mathbb{E}S_n = DS_n = n\lambda$, since the X_i are independent and $\mathbb{E}X_i = DX_i = \lambda$.

2. (0.75 points) If T is the total number of serious accidents in 2021, show $\mathbb{P}(3529 \leq T \leq 3771) \approx 0.954$ and carefully justify the approximation.

   Solution. Use the CLT, explicitly checking its hypotheses, in particular $\mathbb{E}[X_i^2] < \infty$.

Problem 4. Each of $n > 2$ people tosses a fair coin. A person wins if their result differs from everyone else's. If there is no winner, all toss again.

1. (0.25 points) Let X count rounds until a winner. Find its expectation and variance.

   Solution. A winner requires one head and all other tails, or the reverse. This has probability $p_n = n/2^{n-1}$ (why?). Hence $X \sim \text{Ge}(p_n)$, $\mathbb{E}X = 1/p_n$ and $DX = 1/p_n^2$.

2. (0.75 points) After the first winner leaves, continue with the remaining $n-1$ players, then $n-2$, and so on until two remain. What is the expected number of rounds? If winner k receives $100(n-k)$, what is the most you would pay to enter?

   Solution. Explain why the expected number is $1/p_n + 1/p_{n-1} + \cdots + 1/p_3$. By symmetry all players have the same expected winnings, and the total paid is $100[(n-1) + (n-2) + \cdots + 2]$, so divide this total by n.

   Although not required, the total simplifies to $100[n(n-1)/2 - 1]$. The asymptotic behaviour of the other sum is nontrivial and equals $2^n/n$.

Problem 5. A fan collects an album of $n = 500$ different Euro 2020 stickers (in 2021), buying random stickers. Let X be the number needed to collect all types.

1. (0.5 points) Let $X_1 = 1$, and for $i \geq 2$ let X_i count purchases after obtaining the $(i-1)$st distinct sticker through obtaining the ith, inclusive. For example, the sequence $\mathbf{5}, \mathbf{12}, 5, \mathbf{111}, 12, 111, 5, \mathbf{13}$ gives $X_1 = 1, X_2 = 1, X_3 = 2, X_4 = 4$. What is the distribution of X_i? Find $\mathbb{E}X$.

   Solution. $X_2 \sim \mathrm{Ge}(499/500)$, since a new type appears with probability $499/500$; otherwise it is the initial type. Then $X_3 \sim \mathrm{Ge}(498/500)$, and so on. Complete the argument.

2. (0.5 points) Estimate the probability that at least $2n$ purchases are needed.

   Solution. Chebyshev's inequality. This is the well-known coupon collector's problem, also appearing in bonus homework 2. Problems 4 and 5 are in fact very similar.